This TF–IDF score indicates the relative importance of the word *blue* in the given document (the sample sentence) within the context of the specified corpus (eight documents, with *blue* appearing in four of them). Higher TF–IDF scores imply greater importance.

We use TF–IDF here because it's simple to apply and understand. Now that we have the elements represented as vectors, we can measure document similarity using cosine similarity. Cosine similarity is a measure used to calculate the cosine of the angle between two nonzero vectors in a multidimensional space, indicating how similar they are, irrespective of their size.

Figure 8.4 shows how cosine distance compares the vector representations of two pieces or documents of text. Cosine similarity returns a value from –1 (not similar) to 1 (identical). *Cosine distance* is a normalized value ranging from 0 to 2, derived by taking 1 minus the cosine similarity. A cosine distance of 0 means identical items, and 2 indicates complete opposites.

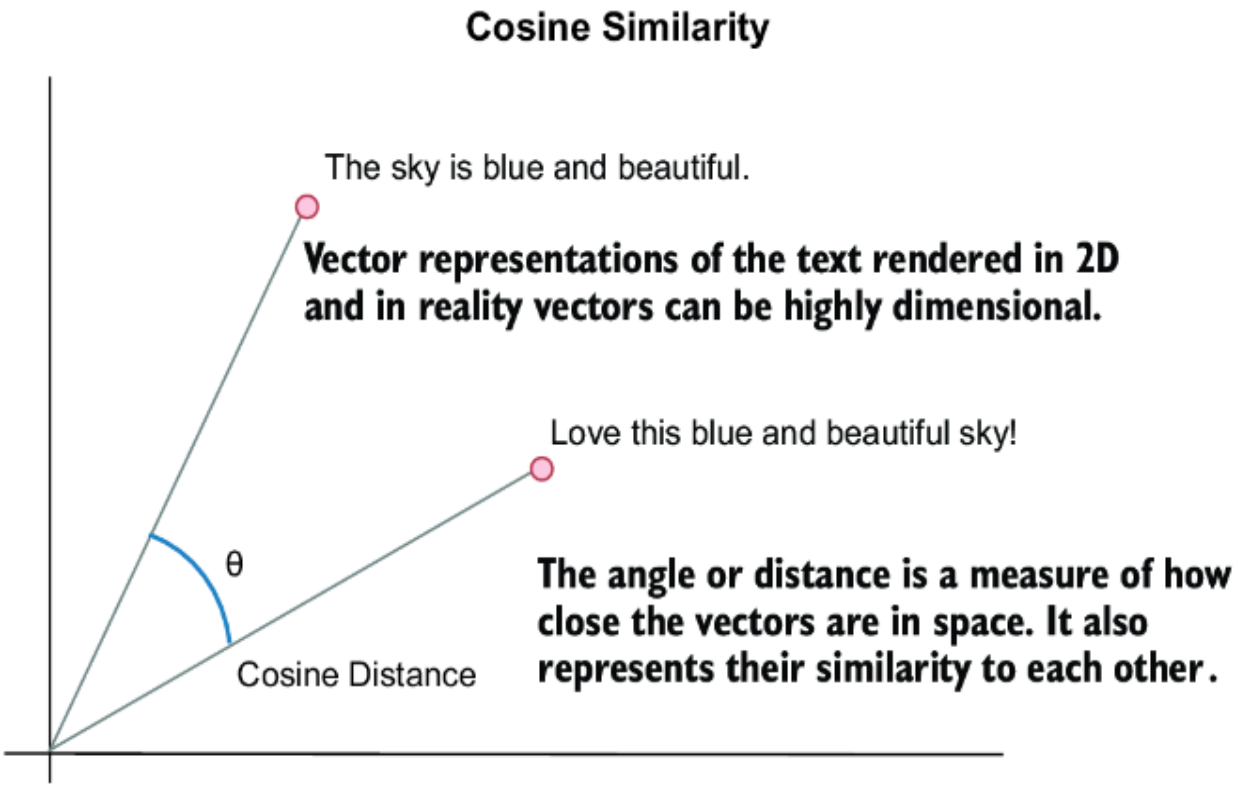


Figure 8.4 How cosine similarity is measured